A Palaeomagnetic Investigation of the Lundy Dyke Swarm

By D. J. BLUNDELL

ABSTRACT

The directions of the natural remanent magnetization of samples collected from dykes on Lundy have been measured and related to those of Tertiary lava flows in Northern Britain. Evidence is given suggesting that the dykes are Tertiary and pre-Upper Oligocene in age, and that the geomagnetic field was reversed at the time of their intrusion.

INTRODUCTION

THE Lundy dyke swarm was shown by Dollar (1941) to have petrological affinities with the Tertiary igneous complexes of Ardnamurchan and Mull, whereas the Lundy granites closely resemble the Armorican granites of South-West England. The dykes are essentially restricted to Lundy and, like the Tertiary dykes in Northern Britain, they have a tendency to strike N.W.–S.E. Only one dyke which would appear to be related to the Lundy dyke swarm is definitely known from the mainland of North Devon, and this outcrops at Lee Bay, near Ilfracombe.

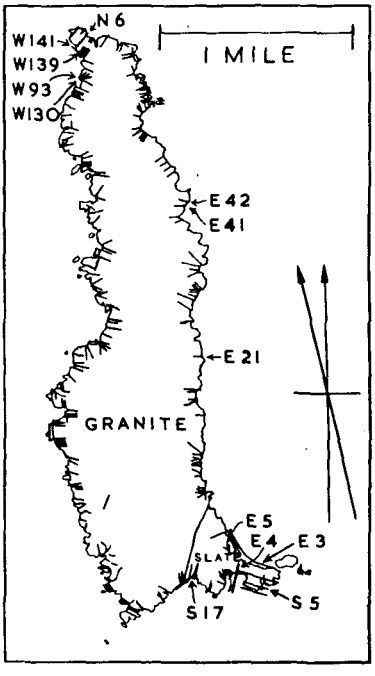
Because of the intimate association of the dykes with the granites on Lundy they have also been regarded (Dollar, 1941, p. 76) as part of the same igneous cycle as the granites, rather than belonging to a separate orogeny. Thus the problem of the age of the dykes is whether they are Permo-Carboniferous or Tertiary. It is felt that rock magnetism is sufficiently advanced to be able to assist in resolving this problem as, in Britain, the directions of remanent magnetization of rocks of these two ages are well separated.

Five oriented samples were collected from the dyke at Lee Bay and, under the guidance of Dr. Dollar, 36 samples were collected from 13 dykes on Lundy, together with five samples of the granites. Although a large number of the 150 dykes mapped by Dollar are inaccessible it was possible to sample reasonably uniformly round the island (Text-fig. 1). Dollar has numbered all the dykes mapped on Lundy, the letters N., S., E., and W. referring to the coast on which they occur, and his notation has been followed. Dykes of various composition and of differing strike directions were sampled. In some cases samples of the baked country rock in contact with the dyke walls were collected, and these proved to be most useful. In all cases the samples were of fresh, unweathered rock. Apart from minor faulting the dykes are unaltered.

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Magnetic Measurements

Two orientated, cubic specimens were cut from each of the samples on which to measure the direction of the remanent magnetization. The strongly magnetized specimens were measured with a spinning magnetometer originally built by Bruckshaw and Robertson (1948) and modified by Vincenz (1952), and the weakly magnetized ones with an astatic magnetometer similar to that



Text-fig. 1.—Lundy Dykes (after Dollar), showing those sampled.

described by Blackett (1952), and capable of measuring magnetic moments of 5×10^{-7} e.m.u. The two instruments were inter-calibrated. The magnetic susceptibility was also measured.

Before measurement every specimen was stored for a month in the present earth's field, oriented as *in situ*. After measurement each one was turned through 180 degrees about an E.–W. axis, to lie for another month, and was then remeasured. This process served two purposes: first, the mean of both measurements removed isothermal and viscous magnetization, leaving only the natural remanent magnetization which was derived from

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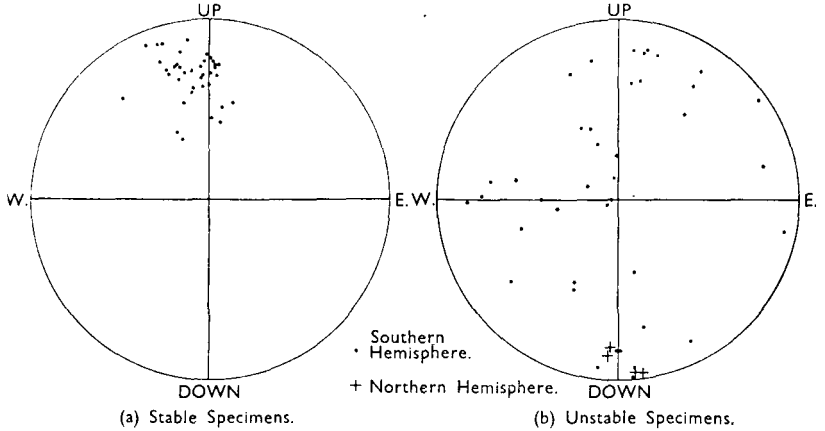
the original thermo-remanent magnetization acquired during the formation of the dyke; second, it is a standard method of testing the stability of the remanent magnetization, since those specimens whose direction of magnetization altered greatly between the two observations are probably unstable.

A further test of stability is afforded by the calculation of the value Q_n, the ratio of natural remanent magnetization to induced magnetization. Dutt (1955) showed that for the Mull lava flows the greater the value of Q_n the greater is the stability of magnetization.

Magnetic Results

Specimens gave varied directions of magnetization according to their degree of stability but, using the two tests mentioned above, they could be divided into stable and unstable groups (Text-fig. 2). The stable specimens are closely grouped together whilst the unstable ones are widely scattered, in agreement with previous work on rock magnetism.

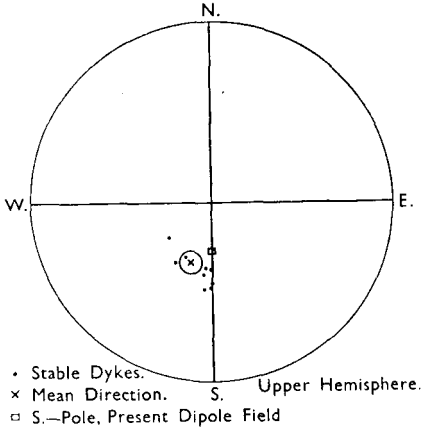
The magnetizations of the stable specimens are opposed in direction to the present earth's field, and those of the unstable ones appear to be the result of an inverse magnetization which has been modified to differing extents by the continued application of the present earth's field. Since various rock compositions are involved this would support the idea of reversal of the geomagnetic field rather than chemically formed inverse magnetization, particularly when the evidence of the baked contact rock is taken into account.



Text-fig. 2.

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Five of the dykes sampled appear to be stable, three partially so, and five are unstable. There is no obvious relation between stability of magnetization and rock composition, as trachytes and dolerites are included in both stable and unstable groups. For an investigation of the age of the dyke swarm the unstable and partially stable dykes must be rejected. In the cases of the dykes E.41 and E.42 (Text-fig. 1), however, samples of contact granite were collected and proved to be magnetically stable so they can be included, even though the dykes themselves are unstable. One



TEXT-FIG. 3.

partially stable dyke, E.5, has a stably magnetized baked slate in contact, and this can also be included.

Table 1 gives the results of the magnetic measurements for the dykes used in the age investigation. The directions of magnetization of these dykes together with their mean value have been plotted on a stereogram, Text-fig. 3. The circle around the mean denotes for the dykes the cone of $33\frac{1}{3}$ per cent confidence, calculated using Fisher's statistical method (Fisher, 1953). It is effectively the spherical equivalent of the standard deviation. The magnetism of Lee Bay dyke is included in Table 1 and Text-fig. 3. The dyke is magnetically stable, and is magnetized exactly as the stable Lundy dykes.

The five samples of granite collected, from each of which two specimens were measured magnetically, were weakly magnetized and the limit of sensitivity of the magnetometer was approached.

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TABLE 1.

Dyke	Rock Composition	Strike	No. of Specimens	Magnetic Intensity ($J \times 10^{-6}$ e.m.u./c.c.)	Q_n	Declination Ø	Dip I
Lundy:—							
E.4	Trachyte*	N. 146° E.	3	9·5	1·0	N. 181° E.	— 60° UP
E.5	Contact Slate (Quartz Dolerite dyke)	80°	2	5·3	0·4	186°	— 60°
E.41	Contact Granite (Dolerite dyke)	108°	2	18·6	1·6	211°	— 58°
E.42	Contact Granite (Dolerite dyke)	133°	2	2·2	0·7	180°	— 53°
S.17	Crinanite	100°	6	2380	1·5	230°	— 65°
W.93	Trachyte	143°	6	12·5	1·9	187°	— 57°
W.141	Crinanite	128°	4	2370	0·9	185°	— 50°
N.6	Trachyte	68°	2	10·8	1·0	206°	— 62°
Lee Bay:—							
	Dolerite	Due E.	10	2540	3·8	181°	— 51°

(* Classified by Dr. Dollar.)

TABLE 2.

Age	Declination Ø	Dip I	Cone of Confidence half angle, α	Formation	
Devonian	N. 198° E.	— 2° UP	5°	Old Red Sandstone, Anglo Welsh Cuvette	(Creer, 1955, p. 73)
Triassic	213°	— 26°	7°	Keuper Marls	(Clegg, Almond, and Stubbs, 1954)
Eocene	194°	— 60°	3°	Antrim Basalts	(Hospers and Charlesworth, 1954)
Miocene?	171°	— 68°	4°	Mull lavas	(Dutt, 1955)
?	197°	— 59°	7°	Lundy dykes	

No conclusive result could be obtained with so few samples, which have a large scatter in directions of magnetization, though they appear to be magnetically stable. The mean direction for the granites has a declination of 175 degrees E. of N. and an upward dip of 9 degrees, with a 27-degree circle of confidence.

Conclusions

A comparison is made in Table 2 between the mean direction of magnetization of the Lundy dykes with those (irrespective of polarity) of British rocks of various ages. The mean direction for the Lundy dykes and the distribution of points about it on the stereogram, Text-fig. 3, closely resemble the result obtained by Hospers and Charlesworth (1954) for the Antrim lavas, which are undoubtedly Tertiary, and probably Eocene, in age. There is also reasonable agreement with the magnetization of the Mull lavas. The direction of magnetization of Carboniferous rocks in this country is still in doubt, but is likely to lie between the directions of Devonian and Triassic rocks, or to have still shallower dip. The magnetization of the Lundy dykes differs by at least 30 degrees from this, and is thus unlikely to be of Permo-Carboniferous age. Its close resemblance to that of the Antrim lavas is strong circumstantial evidence for their being contemporaneous, and supports the suggestion made by Hospers and Charlesworth that there may have been movement of the geographic poles relative to Britain since Eocene times.

The magnetization of the Lee Bay dyke indicates that it is a part of the Lundy swarm. The Lee Bay dyke and some dykes on Lundy are displaced by N.W.–S.E. faults to the same extent as the country rocks. This N.W.–S.E. fault system is regionally developed in South-West England. The Bovey Tracey and Petrockstow Clays appear to have accumulated in rift basins elongated along N.W.–S.E. lines, probably by faulting. The Bovey Clays are generally taken to be of Upper Oligocene age (Reid, 1913) and the faulting would appear to be penecontemporaneous. Hence the relation of the dykes to the faulting dates them as pre-Upper Oligocene.

The magnetization of the Lundy granites is not inconsistent with the idea of their being of Permo-Carboniferous age.

Acknowledgments

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